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## Introduction

### Project Overview

### Business Context

### Project Objective

### Tools & Technology

Methodology

Analytical Framework

Data Collection and Profiing

Data Modeling and Normalization

Analytical Techniques

Statistical Method

Data Schema & Modification

Pre-Normalization Data Schema

Customers

| Column Name              | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|--------------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| customer_id              | text      | 99441           | 99441      | Unique Identifier                 | 100.00%      |
| customer_unique_id       | text      | 96096           | 99441      | Unique Identifier                 | 96.64%       |
| customer_zip_code_prefix | bigint    | 14994           | 99441      | Numerical                         | 15.08%       |
|                          |           |                 |            | Continuous                        |              |
| customer_city            | text      | 4119            | 99441      | Nominal / Text (High Cardinality) | 4.14%        |
| customer_state           | text      | 27              | 99441      | Nominal / Text (High Cardinality) | 0.03%        |

Geolocation

| Column Name                 | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|-----------------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| geolocation_lat             | double    | 717360          | 1000163    | Numerical                         | 71.72%       |
|                             |           |                 |            | Continuous                        |              |
| geolocation_lng             | double    | 717613          | 1000163    | Numerical                         | 71.75%       |
|                             |           |                 |            | Continuous                        |              |
| geolocation_zip_code_prefix | bigint    | 19015           | 1000163    | Numerical                         | 1.90%        |
|                             |           |                 |            | Continuous                        |              |
| geolocation_city            | text      | 5969            | 1000163    | Nominal / Text (High Cardinality) | 0.60%        |

|                   |      |    |         |                                      |       |
|-------------------|------|----|---------|--------------------------------------|-------|
| geolocation_state | text | 27 | 1000163 | Nominal / Text<br>(High Cardinality) | 0.00% |
|-------------------|------|----|---------|--------------------------------------|-------|

▪ Order\_items

| Column Name         | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|---------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| freight_value       | double    | 6999            | 112650     | Numerical<br>Continuous           | 6.21%        |
| order_item_id       | bigint    | 21              | 112650     | Numerical<br>Continuous           | 0.02%        |
| price               | double    | 5968            | 112650     | Numerical<br>Continuous           | 5.30%        |
| order_id            | text      | 98666           | 112650     | Nominal / Text (High Cardinality) | 87.59%       |
| product_id          | text      | 32951           | 112650     | Nominal / Text (High Cardinality) | 29.25%       |
| seller_id           | text      | 3095            | 112650     | Nominal / Text (High Cardinality) | 2.75%        |
| shipping_limit_date | text      | 93318           | 112650     | Nominal / Text (High Cardinality) | 82.84%       |

▪ Order\_payments

| Column Name          | Data Type | Distinct Values | Total Rows | Category                 | Uniqueness % |
|----------------------|-----------|-----------------|------------|--------------------------|--------------|
| order_id             | text      | 99440           | 103886     | Unique Identifier        | 95.72%       |
| payment_installments | bigint    | 24              | 103886     | Numerical<br>Continuous  | 0.02%        |
| payment_sequential   | bigint    | 29              | 103886     | Numerical<br>Continuous  | 0.03%        |
| payment_value        | double    | 29077           | 103886     | Numerical<br>Continuous  | 27.99%       |
| payment_type         | text      | 5               | 103886     | Nominal /<br>Categorical | 0.00%        |

▪ Order\_reviews

| Column Name             | Data Type | Distinct Values | Total Rows | Category                             | Uniqueness % |
|-------------------------|-----------|-----------------|------------|--------------------------------------|--------------|
| order_id                | text      | 98673           | 99224      | Unique Identifier                    | 99.44%       |
| review_answer_timestamp | text      | 98248           | 99224      | Unique Identifier                    | 99.02%       |
| review_id               | text      | 98410           | 99224      | Unique Identifier                    | 99.18%       |
| review_score            | bigint    | 5               | 99224      | Numerical<br>Discrete                | 0.01%        |
| review_comment_message  | text      | 35431           | 99224      | Nominal / Text<br>(High Cardinality) | 35.71%       |
| review_comment_title    | text      | 3901            | 99224      | Nominal / Text<br>(High Cardinality) | 3.93%        |
| review_creation_date    | text      | 636             | 99224      | Nominal / Text<br>(High Cardinality) | 0.64%        |

#### ■ Orders

| Column Name                   | Data Type | Distinct Values | Total Rows | Category                             | Uniqueness % |
|-------------------------------|-----------|-----------------|------------|--------------------------------------|--------------|
| customer_id                   | text      | 99441           | 99441      | Unique Identifier                    | 100.00%      |
| order_delivered_customer_date | text      | 95664           | 99441      | Unique Identifier                    | 96.20%       |
| order_id                      | text      | 99441           | 99441      | Unique Identifier                    | 100.00%      |
| order_purchase_timestamp      | text      | 98875           | 99441      | Unique Identifier                    | 99.43%       |
| order_status                  | text      | 8               | 99441      | Nominal /<br>Categorical             | 0.01%        |
| order_approved_at             | text      | 90733           | 99441      | Nominal / Text<br>(High Cardinality) | 91.24%       |
| order_delivered_carrier_date  | text      | 81018           | 99441      | Nominal / Text<br>(High Cardinality) | 81.47%       |
| order_estimated_delivery_date | text      | 459             | 99441      | Nominal / Text<br>(High Cardinality) | 0.46%        |

#### ■ Product\_category\_name\_translation

| Column Name           | Data Type | Distinct Values | Total Rows | Category          | Uniqueness % |
|-----------------------|-----------|-----------------|------------|-------------------|--------------|
| product_category_name | text      | 71              | 71         | Unique Identifier | 100.00%      |

|                               |      |    |    |                   |         |
|-------------------------------|------|----|----|-------------------|---------|
| product_category_name_english | text | 71 | 71 | Unique Identifier | 100.00% |
|-------------------------------|------|----|----|-------------------|---------|

#### ■ Products

| Column Name                | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|----------------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| product_id                 | text      | 32951           | 32951      | Unique Identifier                 | 100.00%      |
| product_description_lenght | double    | 2960            | 32951      | Numerical Continuous              | 8.98%        |
| product_height_cm          | double    | 102             | 32951      | Numerical Continuous              | 0.31%        |
| product_length_cm          | double    | 99              | 32951      | Numerical Continuous              | 0.30%        |
| product_name_lenght        | double    | 66              | 32951      | Numerical Continuous              | 0.20%        |
| product_weight_g           | double    | 2204            | 32951      | Numerical Continuous              | 6.69%        |
| product_width_cm           | double    | 95              | 32951      | Numerical Continuous              | 0.29%        |
| product_photos_qty         | double    | 19              | 32951      | Numerical Discrete                | 0.06%        |
| product_category_name      | text      | 73              | 32951      | Nominal / Text (High Cardinality) | 0.22%        |

#### ■ Sellers

| Column Name            | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|------------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| seller_id              | text      | 3095            | 3095       | Unique Identifier                 | 100.00%      |
| seller_zip_code_prefix | bigint    | 2246            | 3095       | Numerical Continuous              | 72.57%       |
| seller_city            | text      | 610             | 3095       | Nominal / Text (High Cardinality) | 19.71%       |
| seller_state           | text      | 23              | 3095       | Nominal / Text (High Cardinality) | 0.74%        |

## Table Size in Schema

| Table                             | Size (MB) |
|-----------------------------------|-----------|
| geolocation                       | 66.59     |
| orders                            | 21.55     |
| order_items                       | 20.55     |
| order_reviews                     | 17.55     |
| customers                         | 12.52     |
| order_payments                    | 10.52     |
| products                          | 4.52      |
| sellers                           | 0.28      |
| product_category_name_translation | 0.02      |

## Normalization & Data Modeling

### Data Model Schema

#### ▪ dim\_customer

| Column Name              | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|--------------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| customer_id              | varchar   | 99163           | 99163      | Unique Identifier                 | 100.00%      |
| customer_unique_id       | varchar   | 95828           | 99163      | Unique Identifier                 | 96.64%       |
| customer_city            | varchar   | 4074            | 99163      | Nominal / Text (High Cardinality) | 4.11%        |
| customer_state           | varchar   | 27              | 99163      | Nominal / Text (High Cardinality) | 0.03%        |
| customer_zip_code_prefix | varchar   | 14837           | 99163      | Nominal / Text (High Cardinality) | 14.96%       |

#### ▪ dim\_customer\_geo

| Column Name     | Data Type | Distinct Values | Total Rows | Category             | Uniqueness % |
|-----------------|-----------|-----------------|------------|----------------------|--------------|
| zip_code_prefix | varchar   | 19015           | 19015      | Unique Identifier    | 100.00%      |
| geolocation_lat | decimal   | 18987           | 19015      | Numerical Continuous | 99.85%       |
| geolocation_lng | decimal   | 18988           | 19015      | Numerical Continuous | 99.86%       |

|                   |         |      |       |                                   |        |
|-------------------|---------|------|-------|-----------------------------------|--------|
| geolocation_city  | varchar | 5771 | 19015 | Nominal / Text (High Cardinality) | 30.35% |
| geolocation_state | varchar | 27   | 19015 | Nominal / Text (High Cardinality) | 0.14%  |

▪ dim\_date

| Column Name  | Data Type | Distinct Values | Total Rows | Category              | Uniqueness % |
|--------------|-----------|-----------------|------------|-----------------------|--------------|
| date_key     | int       | 634             | 634        | Numerical Continuous  | 100.00%      |
| day          | int       | 31              | 634        | Numerical Continuous  | 4.89%        |
| week_of_year | int       | 52              | 634        | Numerical Continuous  | 8.20%        |
| is_weekend   | tinyint   | 2               | 634        | Numerical Discrete    | 0.32%        |
| month        | int       | 12              | 634        | Numerical Discrete    | 1.89%        |
| quarter      | int       | 4               | 634        | Numerical Discrete    | 0.63%        |
| year         | int       | 3               | 634        | Numerical Discrete    | 0.47%        |
| full_date    | date      | 634             | 634        | Temporal              | 100.00%      |
| day_name     | varchar   | 7               | 634        | Nominal / Categorical | 1.10%        |
| month_name   | varchar   | 12              | 634        | Nominal / Categorical | 1.89%        |

▪ dim\_order\_payments

| Column Name          | Data Type | Distinct Values | Total Rows | Category             | Uniqueness % |
|----------------------|-----------|-----------------|------------|----------------------|--------------|
| order_id             | varchar   | 99440           | 103886     | Unique Identifier    | 95.72%       |
| payment_id           | int       | 103886          | 103886     | Numerical Continuous | 100.00%      |
| payment_installments | int       | 24              | 103886     | Numerical Continuous | 0.02%        |
| payment_sequential   | int       | 29              | 103886     | Numerical Continuous | 0.03%        |

|               |         |       |        |                          |        |
|---------------|---------|-------|--------|--------------------------|--------|
| payment_value | decimal | 29077 | 103886 | Numerical<br>Continuous  | 27.99% |
| payment_type  | varchar | 5     | 103886 | Nominal /<br>Categorical | 0.00%  |

▪ dim\_order\_review

| Column Name             | Data Type | Distinct<br>Values | Total<br>Rows | Category                             | Uniqueness<br>% |
|-------------------------|-----------|--------------------|---------------|--------------------------------------|-----------------|
| order_id                | varchar   | 98673              | 99224         | Unique Identifier                    | 99.44%          |
| review_id               | varchar   | 98410              | 99224         | Unique Identifier                    | 99.18%          |
| review_score            | int       | 5                  | 99224         | Numerical Discrete                   | 0.01%           |
| review_answer_timestamp | datetime  | 98248              | 99224         | Temporal                             | 99.02%          |
| review_creation_date    | datetime  | 636                | 99224         | Temporal                             | 0.64%           |
| review_comment_message  | text      | 35431              | 99224         | Nominal / Text (High<br>Cardinality) | 35.71%          |
| review_comment_title    | varchar   | 3901               | 99224         | Nominal / Text (High<br>Cardinality) | 3.93%           |

▪ dim\_product

| Column Name                | Data Type | Distinct<br>Values | Total<br>Rows | Category                | Uniqueness<br>% |
|----------------------------|-----------|--------------------|---------------|-------------------------|-----------------|
| product_id                 | varchar   | 2045               | 2045          | Unique<br>Identifier    | 100.00%         |
| product_description_length | int       | 1094               | 2045          | Numerical<br>Continuous | 53.50%          |
| product_height_cm          | decimal   | 77                 | 2045          | Numerical<br>Continuous | 3.77%           |
| product_length_cm          | decimal   | 88                 | 2045          | Numerical<br>Continuous | 4.30%           |
| product_name_length        | int       | 53                 | 2045          | Numerical<br>Continuous | 2.59%           |
| product_weight_g           | decimal   | 474                | 2045          | Numerical<br>Continuous | 23.18%          |
| product_width_cm           | decimal   | 62                 | 2045          | Numerical<br>Continuous | 3.03%           |



|                       |         |    |      |                          |       |
|-----------------------|---------|----|------|--------------------------|-------|
| product_photos_qty    | int     | 16 | 2045 | Numerical<br>Discrete    | 0.78% |
| price_class           | varchar | 0  | 2045 | Nominal /<br>Categorical | 0.00% |
| product_category_name | varchar | 7  | 2045 | Nominal /<br>Categorical | 0.34% |

▪ dim\_product\_category\_name\_translation

| Column Name                   | Data Type | Distinct<br>Values | Total<br>Rows | Category             | Uniqueness<br>% |
|-------------------------------|-----------|--------------------|---------------|----------------------|-----------------|
| product_category_name         | varchar   | 71                 | 71            | Unique<br>Identifier | 100.00%         |
| product_category_name_english | varchar   | 71                 | 71            | Unique<br>Identifier | 100.00%         |

▪ dim\_seller

| Column Name            | Data<br>Type | Distinct<br>Values | Total<br>Rows | Category                             | Uniqueness<br>% |
|------------------------|--------------|--------------------|---------------|--------------------------------------|-----------------|
| seller_id              | varchar      | 3088               | 3088          | Unique Identifier                    | 100.00%         |
| seller_city            | varchar      | 610                | 3088          | Nominal / Text (High<br>Cardinality) | 19.75%          |
| seller_state           | varchar      | 23                 | 3088          | Nominal / Text (High<br>Cardinality) | 0.74%           |
| seller_zip_code_prefix | varchar      | 2239               | 3088          | Nominal / Text (High<br>Cardinality) | 72.51%          |

▪ dim\_seller\_geo

| Column Name      | Data Type | Distinct<br>Values | Total<br>Rows | Category                             | Uniqueness<br>% |
|------------------|-----------|--------------------|---------------|--------------------------------------|-----------------|
| zip_code_prefix  | varchar   | 19015              | 19015         | Unique Identifier                    | 100.00%         |
| geolocation_lat  | decimal   | 18987              | 19015         | Numerical Continuous                 | 99.85%          |
| geolocation_lng  | decimal   | 18988              | 19015         | Numerical Continuous                 | 99.86%          |
| geolocation_city | varchar   | 5771               | 19015         | Nominal / Text (High<br>Cardinality) | 30.35%          |

|                   |         |    |       |                                   |       |
|-------------------|---------|----|-------|-----------------------------------|-------|
| geolocation_state | varchar | 27 | 19015 | Nominal / Text (High Cardinality) | 0.14% |
|-------------------|---------|----|-------|-----------------------------------|-------|

▪ fact\_order\_items

| Column Name         | Data Type | Distinct Values | Total Rows | Category                          | Uniqueness % |
|---------------------|-----------|-----------------|------------|-----------------------------------|--------------|
| freight_value       | decimal   | 6999            | 112650     | Numerical Continuous              | 6.21%        |
| item_id             | int       | 21              | 112650     | Numerical Continuous              | 0.02%        |
| price               | decimal   | 5968            | 112650     | Numerical Continuous              | 5.30%        |
| shipping_limit_date | datetime  | 93318           | 112650     | Temporal                          | 82.84%       |
| order_id            | varchar   | 98666           | 112650     | Nominal / Text (High Cardinality) | 87.59%       |
| product_id          | varchar   | 32951           | 112650     | Nominal / Text (High Cardinality) | 29.25%       |
| seller_id           | varchar   | 3095            | 112650     | Nominal / Text (High Cardinality) | 2.75%        |

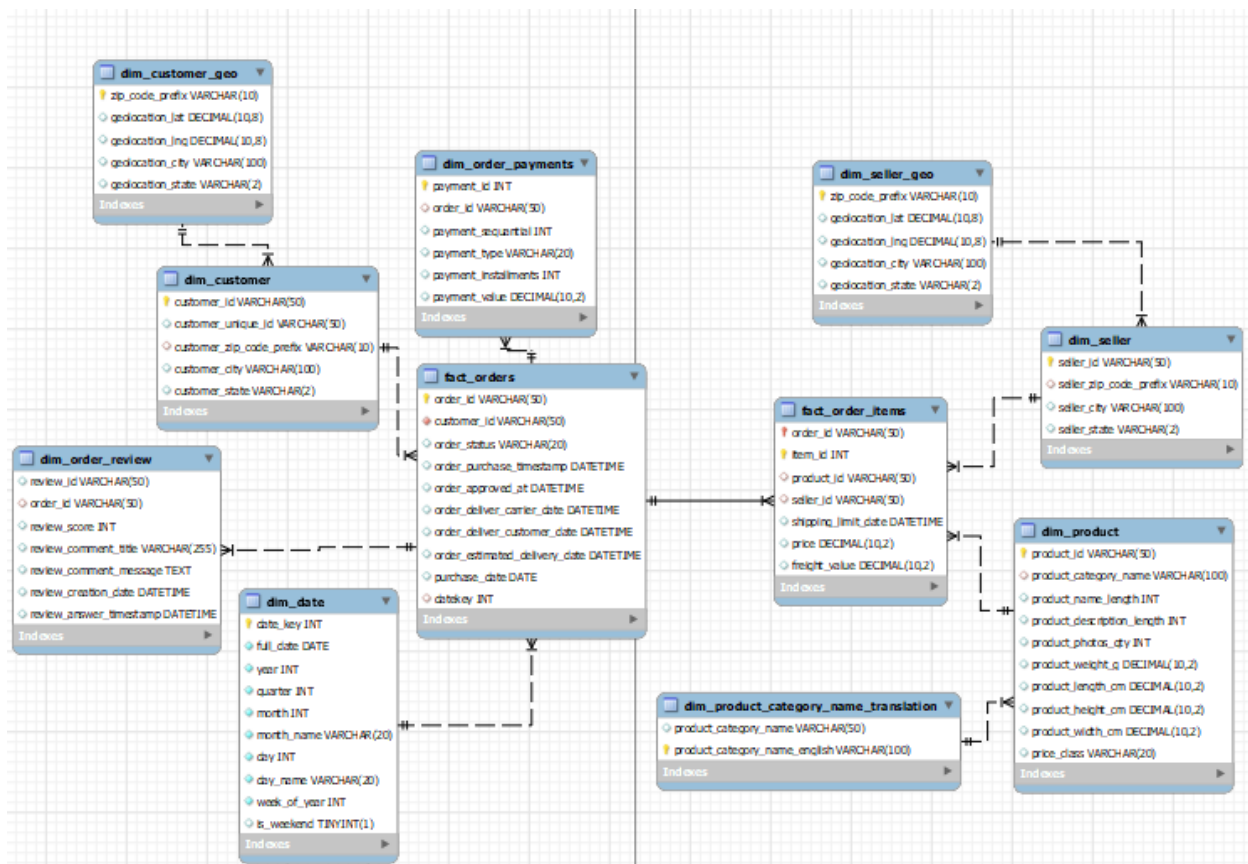
▪ fact\_orders

| Column Name                   | Data Type | Distinct Values | Total Rows | Category              | Uniqueness % |
|-------------------------------|-----------|-----------------|------------|-----------------------|--------------|
| customer_id                   | varchar   | 99163           | 99163      | Unique Identifier     | 100.00%      |
| order_id                      | varchar   | 99163           | 99163      | Unique Identifier     | 100.00%      |
| datekey                       | int       | 634             | 99163      | Numerical Continuous  | 0.64%        |
| order_approved_at             | datetime  | 90498           | 99163      | Temporal              | 91.26%       |
| order_deliver_carrier_date    | datetime  | 80826           | 99163      | Temporal              | 81.51%       |
| order_deliver_customer_date   | datetime  | 95404           | 99163      | Temporal              | 96.21%       |
| order_estimated_delivery_date | datetime  | 459             | 99163      | Temporal              | 0.46%        |
| order_purchase_timestamp      | datetime  | 98601           | 99163      | Temporal              | 99.43%       |
| purchase_date                 | date      | 0               | 99163      | Temporal              | 0.00%        |
| order_status                  | varchar   | 8               | 99163      | Nominal / Categorical | 0.01%        |

- Table size schema

| Table                                 | Size (MB) |
|---------------------------------------|-----------|
| fact_order_items                      | 47.84     |
| fact_orders                           | 32.69     |
| dim_customer                          | 20.11     |
| dim_customer_geo                      | 2.52      |
| dim_seller_geo                        | 2.52      |
| dim_seller                            | 0.52      |
| dim_product                           | 0.39      |
| dim_date                              | 0.06      |
| dim_order_payments                    | 0.03      |
| dim_order_review                      | 0.03      |
| dim_product_category_name_translation | 0.02      |

[ERD Diagram](#)



## Feature Engineering Insight

- Order Approved time
- Delivery Time span
- Carrier handoff time
- Item value
- Total order value
- Freight percentage
- Customer behavior
- Installment category
- Customer location
- Seller performance
- Customer satisfaction
- Customer distance to seller

## Identifier Based Analysis

### Order Identifier Section

#### Order Status

| order_status | order_count | percentage | running_pct |
|--------------|-------------|------------|-------------|
| delivered    | 96214       | 97.03      | 97.03       |
| shipped      | 1100        | 1.11       | 1.11        |
| canceled     | 623         | 0.63       | 0.63        |
| unavailable  | 607         | 0.61       | 0.61        |
| invoiced     | 313         | 0.32       | 0.32        |
| processing   | 300         | 0.30       | 0.30        |
| created      | 4           | 0.00       | 0.00        |
| approved     | 2           | 0.00       | 0.00        |

#### Key Insight

- 97.29% fulfilment industry benchmark
- Very good cancellation rate below then 2-5%
- Operationla flowalso healthy
- Unabilble 0.61%

#### Business Actions

- Investigate root cause of 609 unabailake orders
- Monitortime spent in "Invoice" and "processing" stages

#### Order Level Percentage

| year | quarter | month | month_name | order_count | quarter_avg | pct_of_year |
|------|---------|-------|------------|-------------|-------------|-------------|
| 2016 | 3       | 9     | September  | 4           | 4           | 1.22        |
| 2016 | 4       | 10    | October    | 324         | 163         | 98.48       |
| 2016 | 4       | 12    | December   | 1           | 163         | 0.30        |
| 2017 | 1       | 1     | January    | 800         | 1751        | 1.78        |
| 2017 | 1       | 2     | February   | 1779        | 1751        | 3.96        |
| 2017 | 1       | 3     | March      | 2674        | 1751        | 5.95        |
| 2017 | 2       | 4     | April      | 2398        | 3106        | 5.33        |
| 2017 | 2       | 5     | May        | 3689        | 3106        | 8.20        |
| 2017 | 2       | 6     | June       | 3232        | 3106        | 7.19        |
| 2017 | 3       | 7     | July       | 4017        | 4203        | 8.93        |
| 2017 | 3       | 8     | August     | 4319        | 4203        | 9.60        |
| 2017 | 3       | 9     | September  | 4273        | 4203        | 9.50        |
| 2017 | 4       | 10    | October    | 4617        | 5932        | 10.27       |
| 2017 | 4       | 11    | November   | 7521        | 5932        | 16.72       |
| 2017 | 4       | 12    | December   | 5657        | 5932        | 12.58       |

|      |   |    |           |      |      |       |
|------|---|----|-----------|------|------|-------|
| 2018 | 1 | 1  | January   | 7255 | 7049 | 13.47 |
| 2018 | 1 | 2  | February  | 6704 | 7049 | 12.45 |
| 2018 | 1 | 3  | March     | 7189 | 7049 | 13.35 |
| 2018 | 2 | 4  | April     | 6918 | 6641 | 12.84 |
| 2018 | 2 | 5  | May       | 6855 | 6641 | 12.73 |
| 2018 | 2 | 6  | June      | 6151 | 6641 | 11.42 |
| 2018 | 3 | 7  | July      | 6275 | 4261 | 11.65 |
| 2018 | 3 | 8  | August    | 6491 | 4261 | 12.05 |
| 2018 | 3 | 9  | September | 16   | 4261 | 0.03  |
| 2018 | 4 | 10 | October   | 4    | 4    | 0.01  |

#### Business Insight

- This business showed phenomenal growth in 2017 but faced major disruption in late 2018 that needs urgent investigation .

#### Yearly Growth

| year | Q1 order | Q2 order | Q3 order | Q4 order | Total | YoY Growth % |
|------|----------|----------|----------|----------|-------|--------------|
| 2016 | 0        | 0        | 4        | 325      | 329   | No Data      |
| 2017 | 5253     | 9319     | 12609    | 17795    | 44976 | 13570.52     |
| 2018 | 21148    | 19924    | 12782    | 4        | 53858 | 19.75        |

#### Business Insight

#### Monthly Growth

| year | January | February | March | April | May  | June | July | August | September | October | November | December | total | YOY Growth % |
|------|---------|----------|-------|-------|------|------|------|--------|-----------|---------|----------|----------|-------|--------------|
| 2016 | 0       | 0        | 0     | 0     | 0    | 0    | 0    | 0      | 4         | 324     | 0        | 1        | 329   |              |
| 2017 | 800     | 1779     | 2674  | 2398  | 3689 | 3232 | 4017 | 4319   | 4273      | 4617    | 7521     | 5657     | 44976 | 13570.52     |
| 2018 | 7255    | 6704     | 7189  | 6918  | 6855 | 6151 | 6275 | 6491   | 16        | 4       | 0        | 0        | 53858 | 19.75        |

#### Monthly in Day wise

| month_name | Monday | Tuesday | Wednesday | Thursday | Friday | Saturday | Total |
|------------|--------|---------|-----------|----------|--------|----------|-------|
| January    | 73     | 204     | 225       | 255      | 242    | 220      | 8055  |
| February   | 304    | 278     | 253       | 267      | 346    | 350      | 8483  |
| March      | 375    | 336     | 287       | 316      | 337    | 374      | 9863  |
| April      | 275    | 347     | 318       | 353      | 361    | 284      | 9316  |
| May        | 373    | 423     | 406       | 372      | 307    | 320      | 10544 |
| June       | 308    | 266     | 280       | 326      | 347    | 359      | 9383  |
| July       | 241    | 303     | 333       | 381      | 339    | 275      | 10292 |
| August     | 473    | 457     | 462       | 387      | 391    | 479      | 10810 |
| September  | 157    | 102     | 129       | 151      | 158    | 142      | 4293  |
| October    | 128    | 144     | 207       | 219      | 187    | 179      | 4945  |
| November   | 111    | 124     | 142       | 111      | 142    | 192      | 7521  |
| December   | 274    | 216     | 233       | 335      | 282    | 278      | 5658  |

#### Return Customer

| order_sequence | order_count | customer_count | customer_percentage |
|----------------|-------------|----------------|---------------------|
| First_Order    | 95828       | 95828          | 100.00              |
| Second_Order   | 2988        | 2988           | 3.12                |
| Third_Order    | 251         | 251            | 0.26                |
| 4-5_th_Order   | 68          | 49             | 0.05                |
| 6+_Orders      | 28          | 11             | 0.01                |

#### Days between orders

| repurchase_window | customer_count | avg_days | pct_of_repaet_customers |
|-------------------|----------------|----------|-------------------------|
| 0-30 days         | 1455           | 5.4      | 48.69                   |
| 31-90 days        | 534            | 56.6     | 17.87                   |
| 91-180 days       | 437            | 132.5    | 14.63                   |
| +180 days         | 562            | 292.8    | 18.81                   |

## Customer Identifier Section

### General Information Customer

| total_cust<br>omers | unique_indi<br>viduals | duplicate_custo<br>mer_ids | duplicate_ra<br>te_pct | states_co<br>vered | cities_co<br>vered | zip_codes_<br>overed |
|---------------------|------------------------|----------------------------|------------------------|--------------------|--------------------|----------------------|
| 99163               | 95828                  | 3335                       | 3.36                   | 27                 | 4074               | 14837                |

### Investigate duplicate customer

| ids_per_customer | customer_count | avg_zip_code | pct_of_duplicates |
|------------------|----------------|--------------|-------------------|
| 2                | 2737           | 1.08         | 91.60             |
| 3                | 202            | 1.11         | 6.76              |
| 4                | 30             | 1.07         | 1.00              |
| 5                | 8              | 1.00         | 0.27              |
| 6                | 6              | 1.17         | 0.20              |
| 7                | 3              | 1.00         | 0.10              |
| 9                | 1              | 3.00         | 0.03              |
| 17               | 1              | 1.00         | 0.03              |

### Customer distribution by pareto analysis

| customer<br>_state | customer<br>_count | pct_of_cus<br>tomers | cumulative_<br>customers | cumulative_<br>pct | abc_category       |
|--------------------|--------------------|----------------------|--------------------------|--------------------|--------------------|
| SP                 | 40287              | 42.02                | 40287                    | 42.02              | Pareto A(top 80%)  |
| RJ                 | 12372              | 12.91                | 52659                    | 54.93              | Pareto A(top 80%)  |
| MG                 | 11248              | 11.73                | 63907                    | 66.66              | Pareto A(top 80%)  |
| RS                 | 5273               | 5.50                 | 69180                    | 72.16              | Pareto A(top 80%)  |
| PR                 | 4871               | 5.08                 | 74051                    | 77.24              | Pareto A(top 80%)  |
| SC                 | 3534               | 3.69                 | 77585                    | 80.93              | Pareto B(Next 15%) |
| BA                 | 3268               | 3.41                 | 80853                    | 84.34              | Pareto B(Next 15%) |
| ES                 | 1959               | 2.04                 | 82812                    | 86.38              | Pareto B(Next 15%) |
| GO                 | 1944               | 2.03                 | 84756                    | 88.41              | Pareto B(Next 15%) |
| DF                 | 1908               | 1.99                 | 86664                    | 90.40              | Pareto B(Next 15%) |
| PE                 | 1605               | 1.67                 | 88269                    | 92.07              | Pareto B(Next 15%) |
| CE                 | 1310               | 1.37                 | 89579                    | 93.44              | Pareto B(Next 15%) |
| PA                 | 946                | 0.99                 | 90525                    | 94.43              | Pareto B(Next 15%) |
| MT                 | 875                | 0.91                 | 91400                    | 95.34              | Pareto C(last 5%)  |
| MA                 | 722                | 0.75                 | 92122                    | 96.09              | Pareto C(last 5%)  |
| MS                 | 694                | 0.72                 | 92816                    | 96.82              | Pareto C(last 5%)  |
| PB                 | 517                | 0.54                 | 93333                    | 97.36              | Pareto C(last 5%)  |



|    |     |      |       |        |                   |
|----|-----|------|-------|--------|-------------------|
| PI | 479 | 0.50 | 93812 | 97.86  | Pareto C(last 5%) |
| RN | 472 | 0.49 | 94284 | 98.35  | Pareto C(last 5%) |
| AL | 400 | 0.42 | 94684 | 98.77  | Pareto C(last 5%) |
| SE | 341 | 0.36 | 95025 | 99.12  | Pareto C(last 5%) |
| TO | 272 | 0.28 | 95297 | 99.41  | Pareto C(last 5%) |
| RO | 238 | 0.25 | 95535 | 99.65  | Pareto C(last 5%) |
| AM | 143 | 0.15 | 95678 | 99.80  | Pareto C(last 5%) |
| AC | 77  | 0.08 | 95755 | 99.88  | Pareto C(last 5%) |
| AP | 67  | 0.07 | 95822 | 99.95  | Pareto C(last 5%) |
| RR | 45  | 0.05 | 95867 | 100.00 | Pareto C(last 5%) |

HHI INDEX for market concertation

|           |                        |
|-----------|------------------------|
| hhi_score | concentration_risk     |
| 2173      | Moderate Concentration |

City segmentation with long tail

| city_tier       | city_count | total_customers | avg_customers_per_city | pct_of_customers |
|-----------------|------------|-----------------|------------------------|------------------|
| top 10 citites  | 10         | 33631           | 3363.1                 | 35.05            |
| Top 11-50ciites | 40         | 16961           | 424.0                  | 17.68            |
| Top 51-200      | 150        | 18687           | 124.6                  | 19.48            |
| cities          |            |                 |                        |                  |
| Long Tail(201+) | 4059       | 26671           | 6.6                    | 27.80            |

Top 10 city of customers

| city_rank | customer_city         | customer_count | aszipf_product | moving_avg_zipf |
|-----------|-----------------------|----------------|----------------|-----------------|
| 1         | sao paulo             | 14982          | 14982          | 10234           |
| 2         | rio de janeiro        | 6620           | 13240          | 9927            |
| 3         | belo horizonte        | 2672           | 8016           | 9834            |
| 4         | brasilia              | 1902           | 7608           | 9810            |
| 5         | curitiba              | 1465           | 7325           | 9873            |
| 6         | campinas              | 1398           | 8388           | 9217            |
| 7         | porto alegre          | 1326           | 9282           | 8737            |
| 8         | salvador              | 1205           | 9640           | 8872            |
| 9         | guarulhos             | 1153           | 10377          | 9062            |
| 10        | sao bernardo do campo | 908            | 9080           | 9325            |

# Customer tier revenue pct

| customer_tier | customer_count | pct_of_customers | cumulative_revenue_pct |
|---------------|----------------|------------------|------------------------|
| Top 1%        | 951            | 1.00             | 10.40                  |
| Top 5%        | 4757           | 5.00             | 26.90                  |
| Top 10%       | 9515           | 10.00            | 38.38                  |
| Top 20%       | 19031          | 20.00            | 53.66                  |

# State performance score rate

| customer_state | total_customers | total_orders | total_revenue | avg_order_value | orders_per_customer | delay_rate_pct | avg_review_score | negative_review_pct | quality_score |
|----------------|-----------------|--------------|---------------|-----------------|---------------------|----------------|------------------|---------------------|---------------|
| AP             | 67              | 68           | 16262.80      | 198.33          | 1.015               | 7.35           | 4.22             | 8.96                | 139.78        |
| AC             | 77              | 81           | 19669.70      | 213.80          | 1.052               | 4.94           | 4.10             | 18.75               | 137.79        |
| SP             | 39966           | 41360        | 5949312.62    | 124.71          | 1.035               | 8.93           | 4.13             | 15.96               | 137.08        |
| PR             | 4829            | 4987         | 803445.47     | 139.66          | 1.033               | 7.22           | 4.10             | 16.92               | 136.99        |
| MG             | 11168           | 11534        | 1862639.12    | 141.12          | 1.033               | 7.95           | 4.09             | 16.70               | 136.56        |
| AM             | 142             | 147          | 27918.87      | 168.19          | 1.035               | 6.12           | 4.07             | 20.69               | 136.18        |
| RO             | 233             | 245          | 57295.87      | 207.59          | 1.052               | 6.53           | 4.01             | 19.09               | 135.99        |
| RS             | 5245            | 5428         | 890449.57     | 141.72          | 1.035               | 9.78           | 4.05             | 17.90               | 135.34        |
| RN             | 469             | 480          | 101722.15     | 192.29          | 1.023               | 12.71          | 4.08             | 16.84               | 134.67        |
| TO             | 271             | 278          | 61146.63      | 194.73          | 1.026               | 15.47          | 4.11             | 15.27               | 134.66        |
| MT             | 872             | 901          | 186285.80     | 176.57          | 1.033               | 9.77           | 3.98             | 19.28               | 133.98        |
| DF             | 1897            | 1956         | 326923.73     | 145.75          | 1.031               | 10.74          | 4.00             | 19.17               | 133.94        |
| MS             | 688             | 709          | 137595.08     | 165.58          | 1.031               | 13.68          | 4.05             | 19.09               | 133.79        |
| GO             | 1934            | 1998         | 350449.91     | 149.51          | 1.033               | 11.86          | 3.99             | 18.66               | 133.59        |
| SC             | 3513            | 3612         | 612153.65     | 146.06          | 1.028               | 13.12          | 4.00             | 19.15               | 133.12        |

|    |       |           |                |        |       |       |      |       |            |
|----|-------|-----------|----------------|--------|-------|-------|------|-------|------------|
| ES | 1951  | 2019      | 32547<br>3.07  | 143.82 | 1.035 | 15.01 | 3.99 | 18.12 | 132.7<br>7 |
| PB | 514   | 530       | 13968<br>1.88  | 232.42 | 1.031 | 15.47 | 4.00 | 19.69 | 132.3<br>0 |
| PE | 1601  | 1644      | 32306<br>7.43  | 178.10 | 1.027 | 14.42 | 3.96 | 19.84 | 131.8<br>7 |
| PI | 478   | 490       | 10761<br>0.56  | 199.28 | 1.025 | 20.20 | 3.90 | 20.66 | 128.9<br>9 |
| PA | 941   | 967       | 21766<br>9.04  | 200.80 | 1.028 | 16.13 | 3.79 | 23.50 | 128.1<br>6 |
| BA | 3249  | 3350      | 61239<br>2.55  | 160.86 | 1.031 | 18.51 | 3.81 | 22.88 | 128.0<br>2 |
| RJ | 12291 | 1274<br>9 | 21370<br>45.84 | 145.82 | 1.037 | 17.94 | 3.81 | 26.09 | 127.6<br>1 |
| SE | 337   | 344       | 72891<br>.29   | 189.82 | 1.021 | 20.64 | 3.84 | 24.05 | 127.1<br>9 |
| CE | 1302  | 1323      | 27450<br>3.20  | 186.36 | 1.016 | 20.48 | 3.81 | 23.70 | 126.7<br>8 |
| MA | 715   | 736       | 15051<br>6.59  | 182.44 | 1.029 | 25.00 | 3.72 | 25.86 | 124.0<br>5 |
| AL | 398   | 410       | 96350<br>.47   | 215.55 | 1.030 | 29.51 | 3.71 | 26.96 | 122.3<br>7 |
| RR | 45    | 46        | 10064<br>.62   | 193.55 | 1.022 | 23.91 | 3.58 | 28.26 | 121.4<br>9 |

## Product Identifier Section

Product ,Revenue,Orders

| product_status        | product_count | orders | revenue     |
|-----------------------|---------------|--------|-------------|
| Total Catalog         | 32951         | 0      | 0.00        |
| Active (Sold)         | 32951         | 112650 | 13591643.70 |
| Inactive (Never Sold) | 0             | 0      | 0.00        |

Product performance

| category      | unique_products | total_orders | gross_revenue | total_freight | total_revenue | avg_product_price | avg_freight | avg_total_price | freight_pct_of_price | pct_of_orders | pct_of_revenue |
|---------------|-----------------|--------------|---------------|---------------|---------------|-------------------|-------------|-----------------|----------------------|---------------|----------------|
| Unknown       | 0               | 105017       | 12497450.87   | 2095692.39    | 14593143.26   | 119.00            | 19.96       | 138.96          | 16.77                | 93.22         | 92.11          |
| cool_stuff    | 789             | 3796         | 635290.85     | 84039.10      | 719329.95     | 167.36            | 22.14       | 189.50          | 13.23                | 3.37          | 4.54           |
| pet_hop       | 719             | 1947         | 214315.41     | 39561.24      | 253876.65     | 110.07            | 20.32       | 130.39          | 18.46                | 1.73          | 1.60           |
| console_games | 317             | 1137         | 157465.22     | 19828.02      | 177293.24     | 138.49            | 17.44       | 155.93          | 12.59                | 1.01          | 1.12           |
| audio         | 58              | 364          | 50688.50      | 5710.44       | 56398.94      | 139.25            | 15.69       | 154.94          | 11.27                | 0.32          | 0.36           |
| marketplace   | 104             | 311          | 28378.47      | 5456.06       | 33834.53      | 91.25             | 17.54       | 108.79          | 19.23                | 0.28          | 0.21           |
| dvds_blu_ray  | 48              | 64           | 5999.39       | 1288.74       | 7288.13       | 93.74             | 20.14       | 113.88          | 21.48                | 0.06          | 0.05           |
| la_cuisine    | 10              | 14           | 2054.99       | 333.55        | 2388.54       | 146.79            | 23.83       | 170.61          | 16.23                | 0.01          | 0.02           |

Top 10 product with abc class

| product_id                       | product_category_name | order_count | total_revenue | avg_price | revenue_rank | cumulative_revenue_pct | abc_class           |
|----------------------------------|-----------------------|-------------|---------------|-----------|--------------|------------------------|---------------------|
| bb50f2e236e5eea0100680137654686c |                       | 187         | 67606.10      | 327.62    | 1            | 0.43                   | A (Top 80% Revenue) |
| d1c427060a0f73f6b889a5c7c61f2ac4 |                       | 323         | 60976.03      | 137.65    | 2            | 0.81                   | A (Top 80% Revenue) |
| 6cdd53843498f92890544667809f1595 |                       | 151         | 59093.99      | 350.83    | 3            | 1.18                   | A (Top 80% Revenue) |
| 99a4788cb24856965c36a24e339b6058 |                       | 467         | 51071.60      | 88.17     | 4            | 1.51                   | A (Top 80% Revenue) |
| d6160fb7873f184099d9bc95e30376af |                       | 35          | 50326.18      | 1397.12   | 5            | 1.82                   | A (Top 80% Revenue) |
| 3dd2a17168ec895c781a9191c1e95ad7 |                       | 255         | 48212.22      | 149.94    | 6            | 2.13                   | A (Top 80% Revenue) |
| aca2eb7d00ea1a7b8ebd4e68314663af |                       | 431         | 44820.76      | 71.36     | 7            | 2.41                   | A (Top 80% Revenue) |
| 5f504b3a1c75b73d6151be81eb05bdc9 | cool_stuff            | 63          | 41725.81      | 598.95    | 8            | 2.68                   | A (Top 80% Revenue) |
| 25c38557cf793876c5abdd5931f922db |                       | 38          | 40311.95      | 1023.88   | 9            | 2.93                   | A (Top 80% Revenue) |
| 53b36df67ebb7c41585e8d54d6772e08 |                       | 306         | 39957.93      | 116.67    | 10           | 3.18                   | A (Top 80% Revenue) |

Data Distribution & Visualization

Testing

Overall Analysis

Recommendation

Conclusion